

CRUD OPERATIONS:(MYSQL)

```
package CRUD;
```

```
import java.sql.Connection;
```

```
import java.sql.PreparedStatement;
```

```
import java.sql.ResultSet;
```

```
public class CRUD_Test {
```

```
    public void create_data(String sl_no,String name,String email){
```

```
        DB_Connection obj_DB_Connection=new DB_Connection();
```

```
        Connection connection=obj_DB_Connection.get_connection();
```

```
        PreparedStatement ps=null;
```

```
        try {
```

```
            String query="insert into student(sl_no,name,email) values (?, ?, ?)";
```

```
            ps=connection.prepareStatement(query);
```

```
            ps.setString(1, sl_no);
```

```
            ps.setString(2, name);
```

```
            ps.setString(3, email);
```

```
            System.out.println(ps);
```

```
            ps.executeUpdate();
```

```
        } catch (Exception e) {
```

```
            System.out.println(e);
```

```
        }
```

```
    }
```

```
    public void read_data(String sl_no) {
```

```
        DB_Connection obj_DB_Connection = new DB_Connection();
```

```
        Connection connection = obj_DB_Connection.get_connection();
```

```
        PreparedStatement ps = null;
```

```
        ResultSet rs = null;
```

```
        try {
```

```

String query = "SELECT * FROM student WHERE sl_no = ?";

ps = connection.prepareStatement(query);


ps.setString(1, sl_no);


System.out.println(ps);


rs = ps.executeQuery();


if (rs.next()) {
    System.out.println("Sl no - " + rs.getString("sl_no"));
    System.out.println("Name - " + rs.getString("name"));
    System.out.println("Email - " + rs.getString("email"));
} else {
    System.out.println("No record found for sl_no: " + sl_no);
}


} catch (Exception e) {
    System.out.println(e);
}

}

//

public void update_data(String sl_no,String new_sl_no,String name,String email){
    DB_Connection obj_DB_Connection=new DB_Connection();
    Connection connection=obj_DB_Connection.get_connection();
    PreparedStatement ps=null;
    try {
        String query="update student set sl_no=?,name=?,email=? where
sl_no=?";

```

```

        ps=connection.prepareStatement(query);

        ps.setString(1, new_sl_no);

        ps.setString(2, name);

        ps.setString(3, email);

        ps.setString(4, sl_no);

        System.out.println(ps);

        ps.executeUpdate();
    } catch (Exception e) {

        System.out.println(e);

    }

}

//

public void delete_data(String sl_no){

    DB_Connection obj_DB_Connection=new DB_Connection();

    Connection connection=obj_DB_Connection.get_connection();

    PreparedStatement ps=null;

    try {

        String query="delete from student where sl_no=?";

        ps=connection.prepareStatement(query);

        ps.setString(1, sl_no);

        System.out.println(ps);

        ps.executeUpdate();

    } catch (Exception e) {

        System.out.println(e);

    }

}

public void alter_table() {

    DB_Connection obj_DB_Connection = new DB_Connection();

    Connection connection = obj_DB_Connection.get_connection();

```

```

PreparedStatement ps = null;

try {
    String query = "alter table student ADD City VARCHAR(20)";
    ps = connection.prepareStatement(query);

    System.out.println(ps);

    ps.executeUpdate();
    System.out.println("Column added successfully.");

} catch (Exception e) {
    System.out.println(e);
}

}

public void truncate_data() {
    DB_Connection obj_DB_Connection = new DB_Connection();
    Connection connection = obj_DB_Connection.get_connection();
    PreparedStatement ps = null;

    try {
        String query = "truncate table student";
        ps = connection.prepareStatement(query);

        System.out.println(ps);

        ps.executeUpdate();
        System.out.println("All records deleted successfully.");
    }
}

```

```

    } catch (Exception e) {

        System.out.println(e);

    }

}

public static void main(String[] args) {

    CRUD_Test objTest=new CRUD_Test();

    //insert data

    objTest.create_data("1", "jinu", "jinu@gmail.com");

    objTest.create_data("2", "jawad", "jawad@gmail.com");

    // read data

    objTest.read_data("2");

    //update data

    objTest.update_data("2", "22", "jawad2", "jawad2@gmail");

    //delete data

    objTest.delete_data("1");

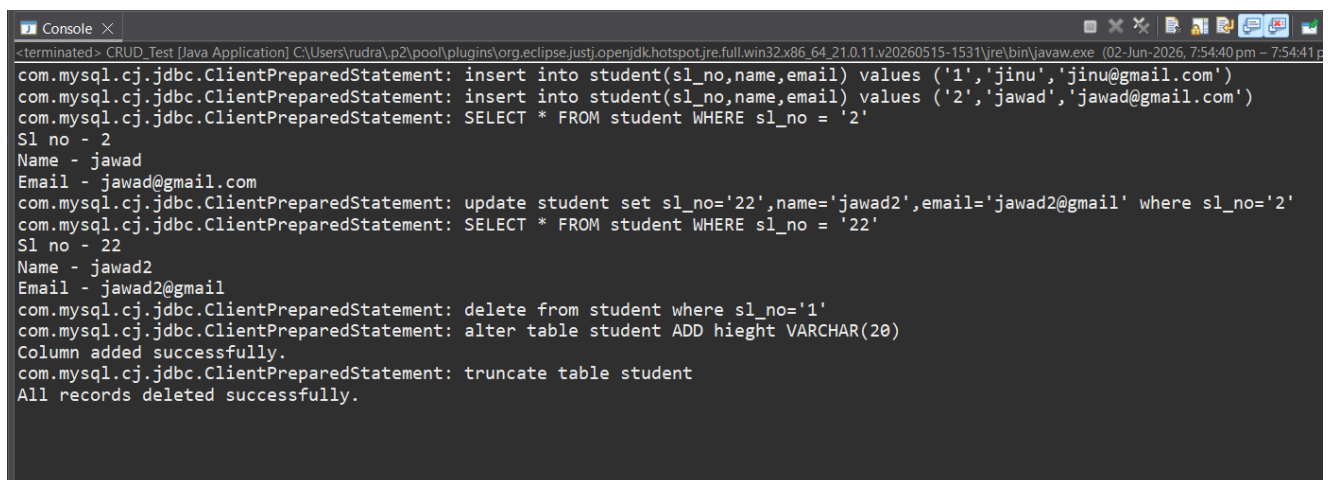
    objTest.alter_table();

    objTest.truncate_data();

}

}

```



The screenshot shows a console window titled "Console" with the following output:

```

<terminated>: CRUD_Test [Java Application] C:\Users\rudra\p2\pool\plugins\org.eclipse.justi.openjdk.hotspot.jre.full.win32.x86_64.21.0.11.v20260515-1531\jre\bin\javaw.exe (02-Jun-2026, 7:54:40 pm - 7:54:41 p
com.mysql.cj.jdbc.ClientPreparedStatement: insert into student(sl_no,name,email) values ('1','jinu','jinu@gmail.com')
com.mysql.cj.jdbc.ClientPreparedStatement: insert into student(sl_no,name,email) values ('2','jawad','jawad@gmail.com')
com.mysql.cj.jdbc.ClientPreparedStatement: SELECT * FROM student WHERE sl_no = '2'
Sl no - 2
Name - jawad
Email - jawad@gmail.com
com.mysql.cj.jdbc.ClientPreparedStatement: update student set sl_no='22',name='jawad2',email='jawad2@gmail' where sl_no='2'
com.mysql.cj.jdbc.ClientPreparedStatement: SELECT * FROM student WHERE sl_no = '22'
Sl no - 22
Name - jawad2
Email - jawad2@gmail
com.mysql.cj.jdbc.ClientPreparedStatement: delete from student where sl_no='1'
com.mysql.cj.jdbc.ClientPreparedStatement: alter table student ADD hieght VARCHAR(20)
Column added successfully.
com.mysql.cj.jdbc.ClientPreparedStatement: truncate table student
All records deleted successfully.

```

MongoDB Connection:

```
package mongodb2;

import com.mongodb.MongoClient;
import com.mongodb.client.MongoDatabase;

public class Test1 {

    public static void main(String[] args) {

        MongoClient mongoClient =

            new MongoClient("localhost", 27017);

        MongoDatabase database =

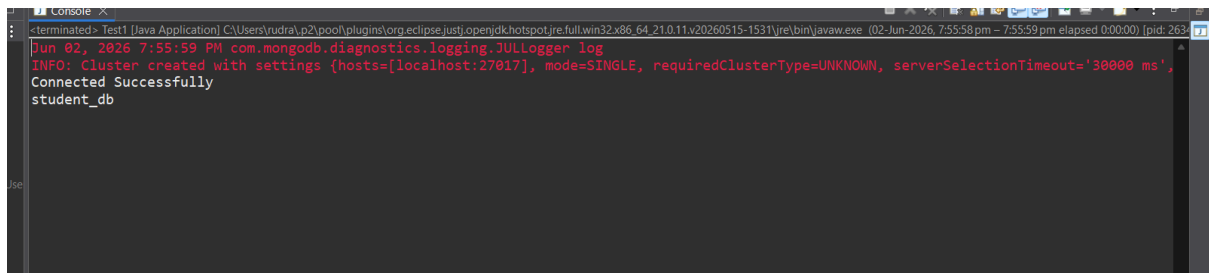
            mongoClient.getDatabase("student_db");

        System.out.println("Connected Successfully");

        System.out.println(database.getName());

    }

}
```

A screenshot of a Java console window titled "Console x". The window shows the output of a Java application. The first line is a red error message: "<terminated> Test1 [Java Application] C:\Users\rudra\p2\poo\plugins\org.eclipse.justi.openjdk.hotspot.jre.full.win32.x86_64.21.0.11.v20260515-1531\jre\bin\javaw.exe (02-Jun-2026, 7:55:58 pm - 7:55:59 pm elapsed 0:00:00) [pid: 263]". The second line is a log message: "Jun 02, 2026 7:55:59 PM com.mongodb.diagnostics.logging.JULLogger log". The third line is an INFO message: "INFO: Cluster created with settings {hosts=[localhost:27017], mode=SINGLE, requiredClusterType=UNKNOWN, serverSelectionTimeout='30000 ms'}". The fourth line is "Connected Successfully". The fifth line is "student_db". The console window has a dark background and a light border.

```
<terminated> Test1 [Java Application] C:\Users\rudra\p2\poo\plugins\org.eclipse.justi.openjdk.hotspot.jre.full.win32.x86_64.21.0.11.v20260515-1531\jre\bin\javaw.exe (02-Jun-2026, 7:55:58 pm - 7:55:59 pm elapsed 0:00:00) [pid: 263]
Jun 02, 2026 7:55:59 PM com.mongodb.diagnostics.logging.JULLogger log
INFO: Cluster created with settings {hosts=[localhost:27017], mode=SINGLE, requiredClusterType=UNKNOWN, serverSelectionTimeout='30000 ms'}
Connected Successfully
student_db
```